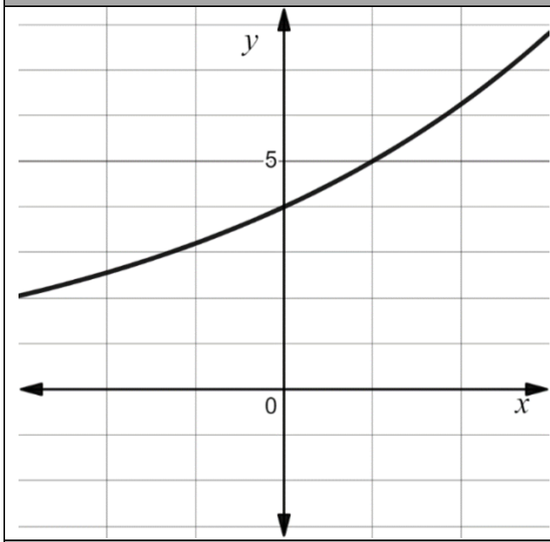
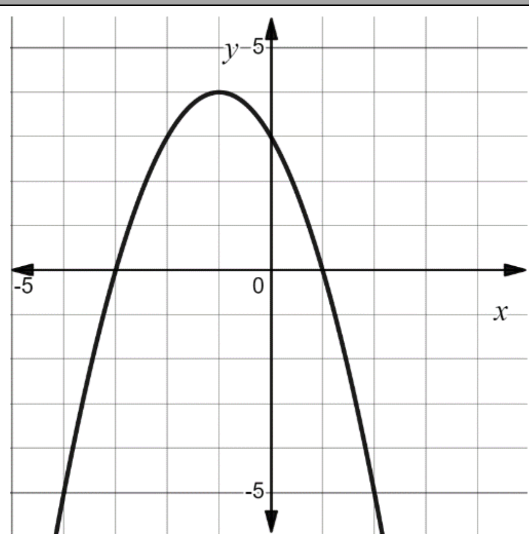
Algebra 1 Name____________________
Unit 13 Assessment Date_________________
Calculator Period_________

1. The table shows six functions.

<table>
<tr><th>Function B</th><th>Function F</th><th>Function H</th></tr>
<tr><td>
</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-6</td><td>-4</td></tr>
<tr><td>-4</td><td>$-\frac{4}{3}$</td></tr>
<tr><td>0</td><td>4</td></tr>
<tr><td>2</td><td>$\frac{20}{3}$</td></tr>
<tr><td>6</td><td>12</td></tr>
</table></td><td>$y = 4(3)^x$</td></tr>
<tr><th>Function M</th><th>Function R</th><th>Function W</th></tr>
<tr><td>
</td><td>$y = 3x^2 + 4$</td><td>$y + 9 = -3(x - 4)$</td></tr>
</table>

Select the functions that share each key feature.

	x –intercept at $(-3, 0)$	x –intercept at $(1, 0)$	y –intercept at $(0,3)$	y –intercept at $(0,4)$
Function B	☐	☐	☐	☐
Function F	☐	☐	☐	☐
Function H	☐	☐	☐	☐
Function M	☐	☐	☐	☐
Function R	☐	☐	☐	☐
Function W	☐	☐	☐	☐

2. In a laboratory experiment, the growth of a tumor in a mouse was monitored over time. The function $T(x) = 0.29(1.14)^x$ models the size of the tumor, where T is the size of the tumor in cubic centimeters (cm^3) and x is the number of days since the tumor was found.

a. Complete the statements.

The function $T(x) = 0.29(1.14)^x$ represents

- ○ linear decay.
- ○ linear growth.
- ○ exponential decay.
- ○ exponential growth.

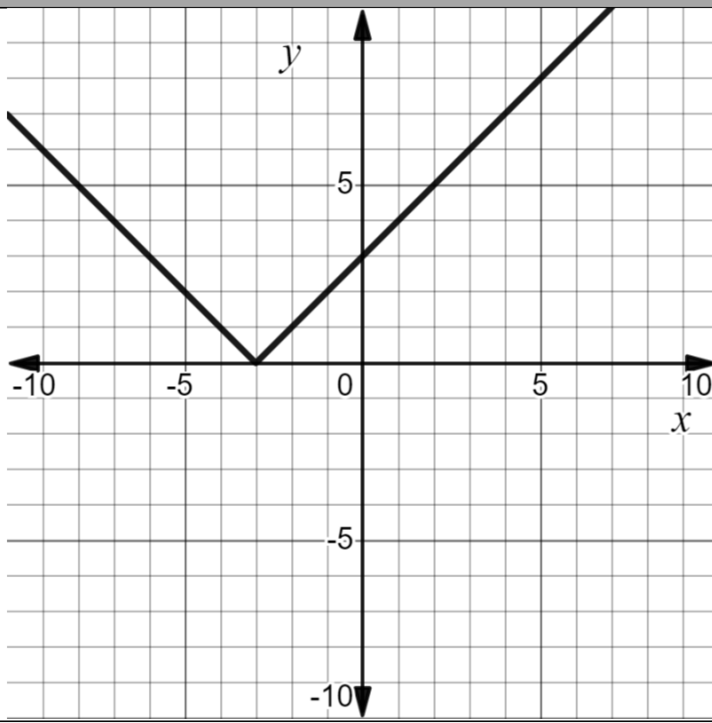
The value 0.29 represents the

- ○ decay rate.
- ○ growth rate.
- ○ the largest size of the tumor, in cm^3.
- ○ the original size of the tumor, in cm^3.

b. Calculate the average rate of change for the interval of day 0 to day 7. Round to the nearest thousandth. Include the units in your answer.

Work	
Average Rate of Change	

3. The table shows a representation of a transformation of a parent function. Determine which family of functions each transformation is a part of by selecting the equation of the parent function.

<table>
<tr><th></th><th>Representation of Transformation</th><th>Equation of the Parent Function</th></tr>
<tr><td>a.</td><td>
</td><td>○ $y = x$
○ $y = x^2$
○ $y = x^3$
○ $y = \sqrt{x}$
○ $y = |x|$
○ $y = 2^x$
○ $y = \frac{1}{2}^x$</td></tr>
<tr><td>b.</td><td><table><tr><th>x</th><th>y</th></tr><tr><td>-5</td><td>-10</td></tr><tr><td>-4</td><td>-8</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>6</td><td>12</td></tr></table></td><td>○ $y = x$
○ $y = x^2$
○ $y = x^3$
○ $y = \sqrt{x}$
○ $y = |x|$
○ $y = 2^x$
○ $y = \frac{1}{2}^x$</td></tr>
</table>

4. Given the parent function $H(x) = x^2$, complete the statements.

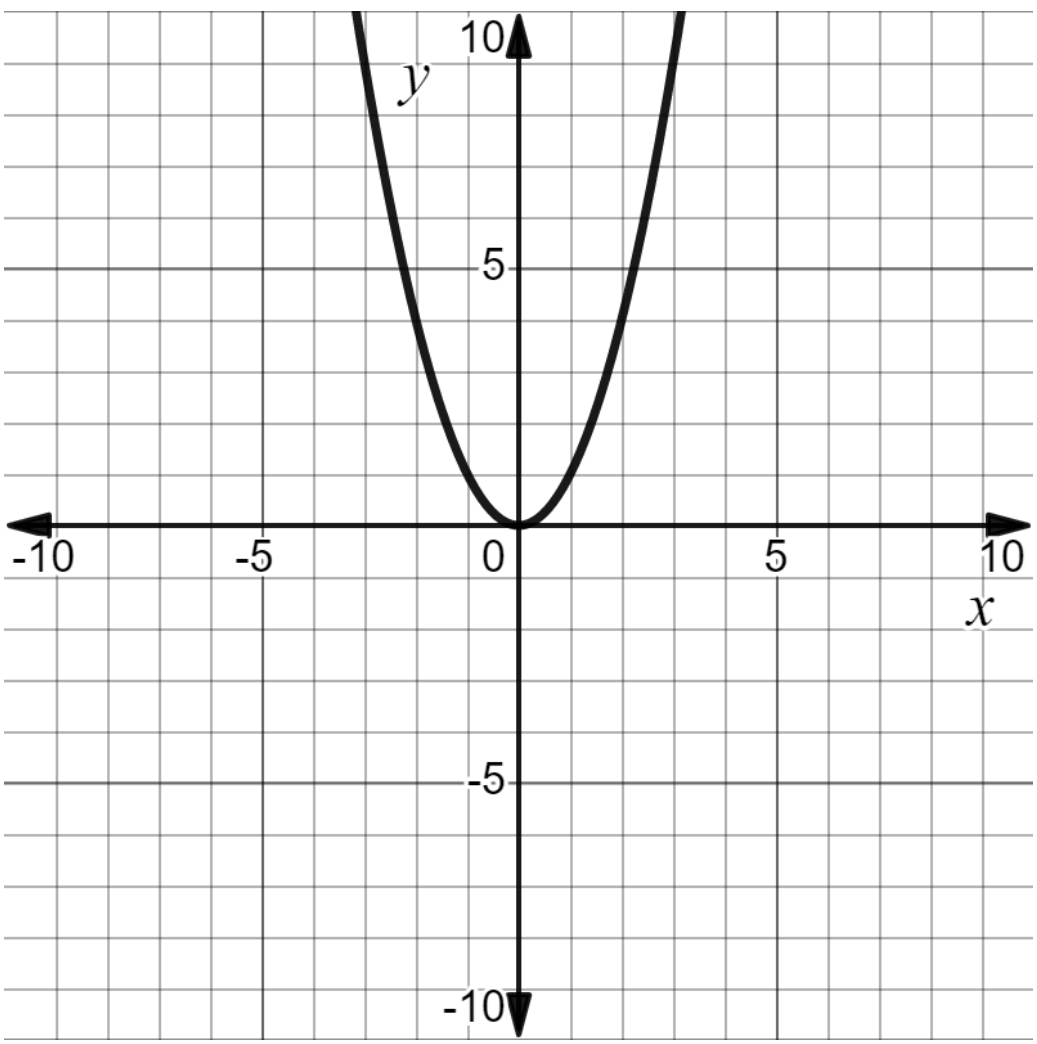
a. The transformation of $H(x)$ represented by $M(x) = -x^2$ can be shown by

- ○ adding -1 to y-values of the function $H(x)$.
- ○ adding -1 to x-values of the function $H(x)$.
- ○ multiplying the y-values of the function $H(x)$ by -1.
- ○ multiplying the x- and the y-values of the function $H(x)$ by -1.

b. The transformation of $H(x)$ represented by $N(x) = (x - 5)^2$ can be shown by

- ○ adding 5 to x-values of the function $H(x)$.
- ○ adding -5 to x-values of the function $H(x)$.
- ○ subtracting 5 from x-values of the function $H(x)$.
- ○ subtracting 25 from x-values of the function $H(x)$.

c. Graph the transformation $H(x)$ represented by $R(x) = x^2 - 4$.



5. When Skylar was born, her parents set up two different savings accounts for her. Each account had an interest rate of 6%. One of the accounts uses simple interest. The other account compounds interest each year. The table shows the amount in each account for the first three years.

Savings Account A		**Savings Account B**	
Years since 2015	**Amount (\$)**	**Years since 2015**	**Amount (\$)**
0	730	0	1,500
1	773.80	1	1,590
2	820.23	2	1,680
3	869.44	3	1,770

a. Complete the statements.

Savings Account B is an example of
- ○ linear
- ○ exponential

growth because
- ○ it has more money.
- ○ the interest earned each year is constant.
- ○ the interest earned each year is compounded.

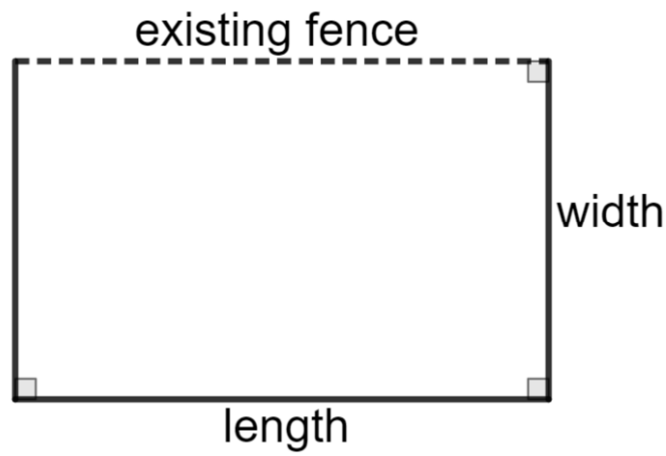
The equation that can be used to determine the amount of money in Savings Account A is
- ○ $A = P(1 + rt)$,
- ○ $A = P\left(1 + \frac{r}{n}\right)^{nt}$,

where A = final amount, P = principal or original amount, t = number of years, and n = number times per year the interest is being compounded, and r = rate of interest per year.

b. Determine the amount in each account after 21 years.

Account	**Work**	**Amount in the Account after 21 years.**
A		
B		

6. A landscape designer is adding a garden to a client's backyard. One side of the garden will be alongside an existing fence. The client has 250 feet of fencing that is left over from another project. The client would like the garden to use all of the fencing. The client also requested the length of the garden should be 4 times the width.



Determine whether a linear, quadratic, or exponential function best models each part of the design.

	Part of the Design	Function Type
a.	Fencing	○ linear ○ quadratic ○ exponential
b.	Area	○ linear ○ quadratic ○ exponential

7. A statistician for an outdoor company analyzes data for the years 2007 to 2014 to help inform the company about trends. The function $R(x) = -14(x-4)^2 + 1000$ models the number of people ages 6 to 17, in tens of thousands, R, who listed running as a hobby for x years since 2007.

Explain what each quantity or expression means within the context.

	Quantity or Expression	Explanation
a.	$(x-4)$	
b.	1000	